Operations Research: transformations drill

Every unusual move from the problem sets and the lecture: to equational form, out of it, minimax and absolute values, free and bounded variables, awkward right-hand sides, and the dual of a non-canonical program

Operations Research, midterm

2026-11-03. Fourteen drills, transformations only (no solving), solution right under each, gotcha at the end.

Where these come from. Problem Set 1 (surplus variables, lower bound by substitution, free inventory variables, the "finish as early as possible" minimax), Problem Set 2 (profit = price − cost, precedence constraints), Problem Set 4 and *How to write a dual* (rows $\geq$ and $=$, variables ≤ 0 and free), the midterm preparation set (Exercise 2 with $x \geq -2$, $y \leq 0$ and a $\geq$ row with negative right-hand side; Exercise 8 with fractions and ratio constraints; Exercise 13 with a negative right-hand side; Exercise 15 with absolute values), and Lecture 2 (the substitution table).

Rules: the whole toolbox on one page

Target: equational form $\max\{c^\top x : Ax = b,\ x \geq 0\}$, $\operatorname{rank} A = m$.

you have	you write	watch out
$a^\top x \leq b$	$a^\top x + s = b$, $s \geq 0$ (slack = unused)	one new column, a unit vector
$a^\top x \geq b$	$a^\top x - e = b$, $e \geq 0$ (surplus = excess)	column $-e_i$: the "all surplus" basis gives $e = -b < 0$, not a BFS
$a^\top x = b$	unchanged	*no* new variable: $q + s = 10$ would allow $q = 7$
$\min c^\top x$	$\max(-c)^\top x$, then negate the value	multipliers change sign too
$x \geq L$	$x = L + u$, $u \geq 0$	a constant appears in every row and in z
$x \leq U$	$x = U - u$, $u \geq 0$	in particular $x \leq 0$: $x = -u$
$L \leq x \leq U$	$x = L + u$, $u \geq 0$, plus the row $u + s = U - L$	one shifted variable, not $u - v$
$x \in \mathbb{R}$ (free)	$x = u - v$, $u, v \geq 0$; or eliminate x with an equation	u and v never both basic: columns are opposite
$b_i < 0$	allowed in equational form	the slack basis is infeasible: pick another starting basis or dual simplex
$\min \max_i(\ell_i(x))$	$\min t$ s.t. $t \geq \ell_i(x)$ for all i	direction: t above every piece
$\max \min_i(\ell_i(x))$	$\max t$ s.t. $t \leq \ell_i(x)$ for all i	direction: t below every piece
$\lvert\ell(x)\rvert \leq \varepsilon$	$\ell(x) \leq \varepsilon$ and $-\ell(x) \leq \varepsilon$	$\lvert\ell(x)\rvert \geq \varepsilon$ is *not* linear (non-convex)
$\min \sum_i \lvert e_i\rvert$	$e_i = u_i - v_i$, $\min \sum(u_i + v_i)$	works because at the optimum one of u_i, v_i is 0
ratio $x_j \leq x_k$, sum $= 1$	move everything left: $x_j - x_k \leq 0$; equality stays	bound rows $x_j \leq 1$ are rows, with their own slack
constant in z	drop it, add it back at the end	it does not change the optimal x or y
scale z by λ	same optimal x	y and reduced costs are multiplied by λ

Dual of a non-canonical maximisation ($\max c^\top x$): row $\leq$ gives $y_i \geq 0$; row $\geq$ gives $y_i \leq 0$; row $=$ gives y_i free; column with $x_j \geq 0$ gives $A_j^\top y \geq c_j$; $x_j \leq 0$ gives $A_j^\top y \leq c_j$; x_j free gives

$A_j^\top y = c_j$; objective $\min b^\top y$. For a **minimisation** primal every rule flips: row $\geq$ gives $y_i \geq 0$, row $\leq$ gives $y_i \leq 0$, $x_j \geq 0$ gives $A_j^\top y \leq c_j$, $x_j \leq 0$ gives $\geq$, free gives $=$; objective $\max b^\top y$.

1 Surplus, not slack (Problem Set 1, Problem 2)

Practice

Given. A ration mixes barley at 5 DH/kg and alfalfa at 8 DH/kg. One kg of barley gives 2 units of protein and 1 of fibre; one kg of alfalfa gives 1 of protein and 3 of fibre. At least 8 units of protein and 9 of fibre are required.

Your task. (a) Write the LP. (b) Convert it to equational form as a maximisation; say what the new variables mean and why they are not slacks. (c) Without solving, explain why the basis formed by the two new variables is not a feasible start for the simplex method. (d) How do you report the value and the multipliers of the original minimisation from those of the maximisation?

Solution

(a) $\min 5x + 8y$ s.t. $2x + y \geq 8$, $x + 3y \geq 9$, $x, y \geq 0$.

(b) $\max -5x - 8y$ s.t. $2x + y - e_1 = 8$, $x + 3y - e_2 = 9$, $x, y, e_1, e_2 \geq 0$. e_1, e_2 are *surplus* variables: protein and fibre supplied beyond the requirement. A slack is subtracted from the stock to reach what is used; here the requirement is a floor, so the excess is *added* to the requirement to reach what is supplied, hence the minus sign in the equation.

(c) Basis (e_1, e_2): $B = -I$, so $x_B = B^{-1}b = -b = (-8, -9)$. The point $(x, y) = (0, 0)$ supplies nothing, so the "all surplus" basis is infeasible whenever $b > 0$. The simplex method needs another starting basis (a hand-picked one, or a phase one).

(d) Value of the original $= -$(value of the max form). Multipliers of the max form are ≤ 0 for $\geq$ rows; negate them to get the marginal costs of the requirements, which are ≥ 0.

Gotcha

Writing $2x + y + s = 8$ for a $\geq$ row is the most common transformation error; it changes the feasible set (it allows $x = y = 0$). The value of the max form is minus the cost; report the cost. The multipliers of the max form are negative; negate them for the minimisation.

2 Lower bound, upper bound, negative right-hand side (prep Exercise 2)

Practice

Given. $\max 4x - 2y$ s.t. $2x - y \leq 10$, $x + y \geq -5$, $x \geq -2$, $y \leq 0$.

Your task. Write an equivalent equational LP. State the substitutions, the new A, b, c, the constant to add back, and how to recover (x, y) from a solution.

Solution

Substitutions. $x \geq -2$: $x = u - 2$, $u \geq 0$. $y \leq 0$: $y = -v$, $v \geq 0$.

Rows. $2x - y \le 10$: $2(u-2) + v \le 10$, i.e. $2u + v \le 14$: $2u + v + s_1 = 14$. $x + y \ge -5$: $u - 2 - v \ge -5$, i.e. $u - v \ge -3$. Keep it as a $\ge$ row with surplus, $u - v - e_2 = -3$, or multiply by -1 first: $-u + v \le 3$, $-u + v + s_2 = 3$. The second choice gives a nonnegative right-hand side and a unit column, so prefer it.

Objective. $4(u-2) - 2(-v) = 4u + 2v - 8$: maximise $4u + 2v$ and subtract 8 at the end.

Result. $\max 4u + 2v$ s.t. $2u + v + s_1 = 14$, $-u + v + s_2 = 3$, $u, v, s_1, s_2 \ge 0$;

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 14 \\ 3 \end{bmatrix}, \quad c = (4, 2, 0, 0)^\top, \quad z_{\text{orig}} = z - 8.$$

Back. $x = u - 2$, $y = -v$, $z_{\text{orig}} = z - 8$.

Gotcha

Do not write $x = u - v$ for $x \ge -2$: one shifted variable is enough and $u - v$ adds a useless column. Do not forget the constant -8 in the objective, and do not forget to substitute in *every* row, including the ones that look untouched.

3 A bounded variable (Lecture 2 table)

Practice

Given. $\max 3t + w$ s.t. $t + w \le 6$, $-2 \le t \le 3$, $w \ge 0$.

Your task. Equational form with the fewest variables. How many variables and equations?

Solution

$t = u - 2$ with $0 \le u \le 5$. The upper bound becomes a row: $u + s_2 = 5$. Row 1: $u - 2 + w \le 6$: $u + w + s_1 = 8$. Objective $3u + w - 6$. Equational form: $\max 3u + w$ s.t. $u + w + s_1 = 8$, $u + s_2 = 5$, all ≥ 0; $n = 4$, $m = 2$; add -6 at the end.

Gotcha

Writing $t = u - v$ and keeping both bounds as rows gives $n = 6$, $m = 3$, and the columns of u and v are opposite, so they can never both be basic; it is correct but slower and the examiner asked for the fewest variables.

4 Free inventory variables (Problem Set 1, Problem 5)

Practice

Given. Two months. Demand 100 then 180; capacity 150 per month; production cost 30 per unit; a unit held one month costs 2; a unit delivered one month late costs a penalty of 5. Inventory I_t at the end of month t may be negative (backlog). The warehouse is empty at the start and everything is settled at the end of month 2.

Your task. (a) Write the balance equations with I_1 free. (b) Make the model linear in nonnegative variables, in two different ways, and write the equational form of one of them. (c) Say why, at an optimum, the holding and backlog parts of I_1 are never both positive.

Solution

(a) $I_1 = p_1 - 100$, $I_2 = I_1 + p_2 - 180 = 0$ (settled), $0 \le p_t \le 150$. Cost $30(p_1+p_2) + 2\max(I_1, 0) + 5\max(-I_1, 0)$: not linear yet.

(b) Way 1, split. $I_1 = h - k$, $h, k \ge 0$ (h held, k backlogged), cost $30p_1 + 30p_2 + 2h + 5k$. Rows: $p_1 - h + k = 100$, $h - k + p_2 = 180$, $p_1 + s_1 = 150$, $p_2 + s_2 = 150$. Equational form (as a max): $\max -30p_1 - 30p_2 - 2h - 5k$ with those four equations, $n = 6$, $m = 4$.
Way 2, eliminate. Use the first balance to remove I_1: $I_1 = p_1 - 100$, so the second row reads $p_1 + p_2 = 280$ and the cost is $30(p_1+p_2) + 2\max(p_1 - 100, 0) + 5\max(100 - p_1, 0)$; then split the one remaining max with $h - k = p_1 - 100$. Same LP, one equation fewer only if you also drop I_1 as a variable.

(c) If $h, k > 0$ both, lowering each by $\min(h, k)$ keeps $h - k$ and every row unchanged and lowers the cost by $7\min(h, k)$. So an optimum has $hk = 0$, which is what makes $h = \max(I_1, 0)$, $k = \max(-I_1, 0)$ legitimate.

Gotcha

The split trick is valid only when the costs of h and k are both *positive* (so that the optimiser has no reason to inflate both). With a negative cost it fails. Also, h and k have opposite columns, so a basis never contains both.

5 Minimax: finish as early as possible (Problem Set 1, Problem 4)

Practice

Given. 6000 posters, three presses at 500, 300, 200 posters per hour in parallel; press C can print at most 1000. Finish time = the slowest press.

Your task. (a) Explain why "minimise the largest of three times" is not linear as stated and fix it with one variable t. (b) Equational form, with n and m. (c) Why is t equal to the largest of the three times at any optimum, although the constraints only say $t \ge$ each time?

Solution

(a) The time is $\max(x_A/500,\ x_B/300,\ x_C/200)$: a maximum of linear functions is piecewise linear, not linear. Introduce t and require t to dominate every piece: $\min t$ s.t. $x_A - 500t \le 0$, $x_B - 300t \le 0$, $x_C - 200t \le 0$, $x_A + x_B + x_C = 6000$, $x_C \le 1000$, $x \ge 0$, $t \ge 0$. At an optimum t equals the largest piece (otherwise lower t).

(b) $\max -t$; slacks s_1, s_2, s_3 on the three press rows, s_4 on the ink row, nothing on the equality: $n = 8$, $m = 5$.

(c) If t were strictly above all three times, decreasing t slightly would keep every constraint and lower the objective; so at an optimum some constraint is tight and $t = \max_i(\cdot)$. The same argument makes the surplus-free reformulation exact.

Gotcha

The direction: for a *min max*, $t \ge$ each piece; for a *max min*, $t \le$ each piece. Writing $t = \max(\dots)$ as an equation is wrong (not linear); the inequality plus the optimisation direction does the job.

6 Maximin

Practice

Given. Two customers value a bundle (x, y) at $3x+y$ and $x+4y$; you can offer at most $x+y \le 10$, $x, y \ge 0$, and want to maximise the *smaller* of the two valuations.

Your task. Write the LP and its equational form. Why must both valuations be equal at an optimum (no computation)?

Solution

$\max t$ s.t. $t \le 3x+y$, $t \le x+4y$, $x+y \le 10$. Rows: $-3x-y+t+s_1 = 0$, $-x-4y+t+s_2 = 0$, $x+y+s_3 = 10$; $n = 6$, $m = 3$. At an optimum t equals the smaller valuation (otherwise raise t). If the two valuations differed, moving a little of the bundle towards the smaller one would raise the minimum, so at an optimum they are equal, unless a bound stops the move.

7 Absolute values (midterm prep, Exercise 15)

Practice

Given. Points $(t_i, y_i) = (0,1), (1,4), (2,2)$. Fit a line $a+bt$ with $a, b \in \mathbb{R}$.

Your task. (a) Write the LP for $\min_{a,b} \max_i |a+bt_i - y_i|$ with no max and no $|\cdot|$. (b) Write the LP for $\min_{a,b} \sum_i |a+bt_i - y_i|$. (c) Explain why "$|a+bt_i - y_i| \ge 1$ for all i" cannot be written as linear constraints. (d) Put (a) in equational form: substitutions, n, m.

Solution

(a) $\min \varepsilon$ s.t. $a+bt_i - y_i \le \varepsilon$ and $-(a+bt_i - y_i) \le \varepsilon$ for $i = 1,2,3$, a, b free, $\varepsilon \ge 0$: six rows. Explicitly $a-1 \le \varepsilon$, $1-a \le \varepsilon$, $a+b-4 \le \varepsilon$, $4-a-b \le \varepsilon$, $a+2b-2 \le \varepsilon$, $2-a-2b \le \varepsilon$.

(b) $e_i = a+bt_i - y_i = u_i - v_i$, $u_i, v_i \ge 0$: $\min \sum_i (u_i+v_i)$ s.t. $a+bt_i - u_i + v_i = y_i$, a, b free. Three equality rows, no slack.

(c) $|e| \ge 1$ means $e \ge 1$ *or* $e \le -1$: the union of two half-lines, not convex, so no system of linear inequalities describes it. $|e| \le 1$ is an intersection, hence fine.

(d) $a = a^+ - a^-$, $b = b^+ - b^-$ (genuinely free), $\varepsilon \ge 0$ already, slack on each of the six rows: $n = 5+6 = 11$, $m = 6$. Objective $\max -\varepsilon$.

Gotcha

"$\le \varepsilon$ on both sides" is two rows per point, not one. In (b) do not write $|e_i| = u_i + v_i$ as a constraint; it is the objective that forces one of u_i, v_i to zero.

8 Fractions, ratio and sum constraints (midterm prep, Exercise 8)

Practice

Given. Fractions $x_A, \dots, x_E \in [0,1]$ of five projects are funded. Budget $4x_A + 6x_B + 5x_C + 3x_D + 2x_E \le 14$, staff $2x_A + 3x_B + 2x_C + 2x_D + x_E \le 7$; C cannot exceed A; B and D sum to 1; E cannot exceed $B + C$.

Your task. Write the constraints as rows ready for the equational form, say which rows get a slack, and count n and m.

Solution

Rows: budget ≤ 14 (slack), staff ≤ 7 (slack), $x_C - x_A \leq 0$ (slack), $x_B + x_D = 1$ (no slack), $x_E - x_B - x_C \leq 0$ (slack), and the five upper bounds $x_j \leq 1$ (five slacks). $x \geq 0$ is the sign condition, not a row. $n = 5 + 9 = 14$, $m = 10$. Objective $\max 8x_A + 12x_B + 11x_C + 6x_D + 5x_E$.

Gotcha

A ratio constraint must be moved to one side: "$x_C \leq x_A$" is the row $x_C - x_A \leq 0$ with right-hand side 0 (a degenerate-looking row, which is normal). Bounds $x_j \leq 1$ are rows with their own slacks; forgetting them changes m.

9 Negative right-hand side and the starting basis (midterm prep, Exercise 13; Problem Set 4)

Practice

Given. $\max 2x_1 + x_2$ s.t. $-x_1 + x_2 \leq -1$, $x_1 + x_2 \leq 5$, $x \geq 0$.

Your task. (a) Equational form. (b) Show that the slack basis is not feasible. (c) Find a feasible starting basis by hand and write its tableau (do not solve further).

Solution

(a) $-x_1 + x_2 + s_1 = -1$, $x_1 + x_2 + s_2 = 5$. Equational form accepts $b_1 = -1$; rank is 2 (slack columns).

(b) Slack basis: $s_1 = -1 < 0$. Multiplying row 1 by -1 gives $x_1 - x_2 - s_1 = 1$ with a nonnegative right-hand side but no unit column for s_1 any more, so it does not help by itself.

(c) In row 1 pick a variable with a *positive* coefficient after the sign flip: x_1. Basis (x_1, s_2): from row 1, $x_1 = 1 + x_2 + s_1$; substitute in row 2: $s_2 = 4 - 2x_2 - s_1$; $z = 2 + 3x_2 + 2s_1$. Constants $1, 4 \geq 0$: a BFS, $x = (1, 0)$.

Gotcha

A negative b_i is not an error in the equational form; it only kills the slack start. The dual simplex is the other way out: keep the slack basis (reduced costs $2, 1 > 0$ there, so it is not dual feasible either, which is why a hand-picked basis is the right tool here).

10 Dual of a non-canonical minimisation (Problem Set 4 and *How to write a dual*)

Practice

Given. $\min 4x_1 + 2x_2 - x_3$ s.t. $x_1 - x_2 + x_3 \geq 3$, $2x_1 + x_3 \leq 8$, $x_1 + x_2 = 4$, $x_1 \geq 0$, $x_2 \in \mathbb{R}$, $x_3 \leq 0$.

Your task. (a) Write the dual directly with the sign table for a minimisation. (b) Check it by

converting the primal to a maximisation first and dualising that. (c) Which dual variable is free, and why?

Solution

(a) Dual variables w_1, w_2, w_3 for the three rows; minimisation primal: row $\geq$ gives $w_1 \geq 0$, row $\leq$ gives $w_2 \leq 0$, row $=$ gives w_3 free. Columns: $x_1 \geq 0$ gives $\leq c_1$; x_2 free gives $= c_2$; $x_3 \leq 0$ gives $\geq c_3$.

$$\max 3w_1 + 8w_2 + 4w_3 \quad \text{s.t.} \quad w_1 + 2w_2 + w_3 \leq 4, \quad -w_1 + w_3 = 2, \quad w_1 + w_2 \geq -1,$$
$$w_1 \geq 0, \; w_2 \leq 0, \; w_3 \in \mathbb{R}.$$

(b) Primal as $\max -4x_1 - 2x_2 + x_3$. Maximisation rules: row $\geq$ gives $y_1 \leq 0$, row $\leq$ gives $y_2 \geq 0$, row $=$ gives y_3 free; $x_1 \geq 0$ gives $A_1^\top y \geq -4$: $y_1 + 2y_2 + y_3 \geq -4$; x_2 free: $-y_1 + y_3 = -2$; $x_3 \leq 0$: $y_1 + y_2 \leq 1$; objective $\min 3y_1 + 8y_2 + 4y_3$. Put $y = -w$: $w_1 \geq 0$, $w_2 \leq 0$, $w_1 + 2w_2 + w_3 \leq 4$, $-w_1 + w_3 = 2$, $w_1 + w_2 \geq -1$, objective $\min -(3w_1 + 8w_2 + 4w_3) = -\max(3w_1 + 8w_2 + 4w_3)$. Same dual as (a), and the two optimal values agree after the sign change.

(c) w_3, attached to the equality row: an equality can be multiplied by any real number and still be used in a bound, so its multiplier has no sign.

Gotcha

Never multiply a row by -1 *and* also apply the "$\geq$ row gives $y \leq 0$" rule: that flips twice. Either convert the row and use the canonical rule, or keep the row and use the sign rule, not both. The sign conditions $x_j \geq 0$ are not rows; they decide the *direction* of the dual constraints.

11 Back to the original variables (Problem Set 1, Problem 3)

Practice

Given. 100 ha: wheat x_1, sunflower x_2, fallow x_3, every hectare used ($x_1 + x_2 + x_3 = 100$). Water $3x_1 + 2x_2 \leq 240$. Contract $x_1 \geq 20$. Margins $4, 5, 1$ (thousand DH/ha).

Your task. (a) Substitute away the lower bound and write the equational form. (b) Suppose the transformed program has been solved and u is nonbasic at 0, x_2 and the water slack are basic. Without numbers: how do you recover the original allocation, the margin, and which constraints bind?

Solution

(a) $x_1 = 20 + u$. Land: $u + x_2 + x_3 = 80$ (equality, no slack). Water: $3u + 2x_2 + s = 180$. Objective $4u + 5x_2 + x_3 + 80$. Equational form: $\max 4u + 5x_2 + x_3$ s.t. the two rows, all variables ≥ 0; $n = 4$, $m = 2$.

(b) $x_1 = 20 + u$ gives the wheat area; x_2, x_3 are unchanged; margin $= z + 80$. Land is binding by construction (equality). Water is not binding since its slack is basic and positive. The contract is binding because $u = 0$: the bound $x_1 \geq 20$ is active, and the reduced cost of u is its shadow price (how much margin one more forced hectare of wheat costs).

Gotcha

After a substitution, "u nonbasic at 0" means "the bound is active", and the reduced cost of u is the shadow price of the bound. Report the answer in the original variables and add the constant back.

12 Profit, cost and scaling (Problem Set 2, Problem 1)

Practice

Given. Blend A sells at 900 MAD and costs 400 per unit; blend B sells at 700 and costs 300. Processing $2x_A + x_B \le 10$, packaging $x_A + 2x_B \le 8$.

Your task. (a) Write the objective in hundreds of MAD. (b) If the scaled program's optimal tableau shows multipliers y, what are the shadow prices in MAD per hour? (c) Does scaling change the optimal plan?

Solution

(a) Profit per unit = price − cost: 500 and 400 MAD, i.e. $\max 5x_A + 4x_B$ in hundreds.

(b) The multipliers scale with the objective: multiply y by 100 to get MAD per hour.

(c) No: multiplying c by $\lambda > 0$ multiplies every reduced cost by λ, so the sign tests, hence the optimal basis and plan, are unchanged; $z^\star$ and y are multiplied by λ.

13 Precedence and deadlines (Problem Set 2, Problem 3)

Practice

Given. Activities A (6 days) and B (5 days) start at day 0 in parallel; C (4 days) starts after both finish. Each can be shortened by $x \in [0, 2]$ days at $4, 3, 5$ thousand MAD per day. Deadline: day 8.

Your task. Formulate the cheapest crash plan as an LP without any max, and give its equational size.

Solution

Let S_C be the start of C. "After both finish" is two inequalities: $S_C \ge 6 - x_A$, $S_C \ge 5 - x_B$. Deadline: $S_C + 4 - x_C \le 8$. Bounds $0 \le x_A, x_B, x_C \le 2$, $S_C \ge 0$.

$$\min 4x_A + 3x_B + 5x_C \quad \text{s.t.} \quad x_A + S_C \ge 6, \quad x_B + S_C \ge 5, \quad -x_C + S_C \le 4,$$
$$x_A, x_B, x_C \le 2, \quad x, S_C \ge 0.$$

Equational form: surplus on the two $\ge$ rows, slack on the four $\le$ rows: $n = 4 + 6 = 10$, $m = 6$, as a max of $-4x_A - 3x_B - 5x_C$.

Gotcha

"Starts after both finish" is a *max* in words but two $\ge$ rows in the model, with a common event variable S_C. Shortening bounds $x \le 2$ are rows (or substitute $x = 2 - u$ if you prefer fewer rows but then $u \ge 0$ and the cost becomes $4(2 - u)$ with a constant).

14 Equality rows and the temptation to add a variable (Lecture 2)

Practice

Given. A contract says exactly 10 tonnes must be delivered: $q_1 + q_2 = 10$.

Your task. Explain why neither $q_1 + q_2 + s = 10$ nor $q_1 + q_2 - s = 10$ with $s \geq 0$ is acceptable, and how an equality row behaves in a basis.

Solution

$q_1 + q_2 + s = 10$ allows $q_1 + q_2 = 7$, $s = 3$: delivering less. $q_1 + q_2 - s = 10$ allows 13 with $s = 3$: delivering more. Both change the feasible set. The row stays as it is. In the basis it still counts as one equation (m includes it), so one basic variable is "consumed" by it; the slack start is impossible when there is an equality row, which is why the problem sets give you a BFS to start from or ask you to find one by hand (drill 9).

Gotcha

Counting m: one per row, including equalities, after any bound rows are added. Counting n: original variables after substitution, plus one slack or surplus per inequality row, plus the u, v pairs of genuinely free variables.